

Precedence — Era Unlock Map (design lock draft)

Math Sensei · 3 Oct 2026 · for Julius before build

Working title: **Precedence**. Scope of this doc: the **order-of-operations racer** only (ten eras of getting around). Algebra Licenses 2–11 stay a later wing unless noted.

1. Design thesis

Vehicle = rank. Math = the gate. Road world = the souvenir.

You never buy a car. You earn how you move. New players always start **on foot**. The Miami wedge / 80s burnout is **Era 8**, not the default skin and not the attract-screen spoiler on a fresh install.

Each era adds **one new rule** (or one sharp combination). Clearing the era unlocks:

1. the next vehicle sprite on the road
2. that era's vista / lamps / sky pack
3. that era's "engine" sound (footsteps → hooves → clatter → rumble → neon whine)
4. the garage postcard stamped **CLEARED**

Practice has no clock. Rally / Race is where speed and the vehicle fantasy matter. A miss costs time (and a hubcap once cars exist), never a game-over scare.

2. The ten eras (unlock table)

Era	Vehicle	Place / vibe	New math rule	Example	Suggested Mathera skills	Earn condition (MVP proposal)
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1	On foot	c. 1200 · Mesa Verde cliffs	· '+' and '-' only; brackets from day one	· $9 - 4 + 3$, · $8 + (5 + 2)$	· II.8.01, III.4.03 · Stamp Practice ×10 clean	· **or** clear Rally checkpoint pack
2	Royal litter	· 40 BC · Cleopatra / Alexandria	· adds 'x' (x before +-)	· $2 + 3 \times 4$	· II.8.01, II.8.03, III.4.03	· Same
3	Horseback	· 1860s · desert West	· brackets change order	· $(2 + 3) \times 4$	· II.8.01, II.8.03, III.4.03	· Same
4	Carriage	· 1850s · gaslit Boston	· adds '÷'; 'x÷' left to right	· $12 \div 3 \times 2$	· II.8.01, II.8.03, III.4.03	· Same
5	Model T	· 1913 · Detroit line	· mixes all four freely	· $18 - 6 \div 2 + 1$	· II.8.01, II.8.03, III.4.03	· Same + short **road test** (12 mixed, 1 slip OK) → unlock Era 6
6	Streamliner	· 1935 · Art Deco NY	· adds powers	· $2 + 3^2$	· II.8.01, II.8.03, III.1.08?	· III.4.03 · Same
7	Fifties cruiser	· 1957 · Route 66 diner	· brackets with powers	· $(1 + 2)^2 \times 2$	· same family	· Same
8	**Wedge sports car**	· 1986 · Miami sunset	· nested brackets	· $2 \times (3 + (4 - 1))$	· same family	· **Burnout unlock** – feel this as the fantasy payoff
9	Electric	· Today · SF fog	· negatives	· $-3 + 4 \times (-2)$	· extend Ops / integers	· Same
10	Robotaxi	· 2040 · neon city	· fraction bars + everything	· compound	· hand-off toward Fractions app / III fraction ops	· Elite plate; optional Pro ★

Notes on chronology: eras are ordered by *how you travel*, not strict history. Horse (1860s) before Carriage (1850s) is intentional fantasy rank, not a textbook timeline. Keep the postcard copy playful, not a museum lecture.

Notes on engines today: engine.js only has levels 1–6 (Sums → Long chains) and POWERS = false in the live generator. This map **stretches** that into 10 eras by splitting "same ops, harder structure" (eras 3, 5, 7, 8) and adding negatives / fraction bars late. Build work = remap levels + turn powers on per era + new sprites/worlds.

3. Progression model (detail)

3.1 What "clearing an era" means

- **Practice:** earn stamps on that era's level(s). Proposal: **10 clean problems** → **1 stamp**; **5 stamps** → **era cleared** (reuse Algebra Licenses stamp math so one save format works).

- **Rally**: optional for clearing; required for **best km** and for feeling the vehicle. Streaks raise speed; miss = -10 s (or -5 s if we keep v3.1 pacing).
- **Road test (eras 5 and 10)**: mixed problems at top tier; pass unlocks the next *band* (pre-car → cars; cars → elite).

3.2 What unlocks visually

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Band · Eras · Road fantasy
|---|---|---|
Pre-car · 1-4 · Dirt / stone / sand paths; camera still "pseudo-3D road" but vehicle is walker / litter
/ horse / carriage
Early car · 5-7 · Real car silhouette evolves: boxy T → deco streamliner → chrome cruiser
Fantasy car · **8** · Miami wedge – neon, hubcaps, burnout audio sting on streak ≥5
Future · 9-10 · Quiet EV hum → robotaxi HUD lanes

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Important: on a fresh profile, attract / title can show a **generic dusk road** or Era 1 footpath — **not** Miami. Garage shows locked silhouettes as outlines ("??") until earned.

3.3 Garage UI (replace dense plate strip for Precedence MVP)

Horizontal or 2×5 postcard garage (from concept art):

- Locked: grey outline + math tease ("adds x")
- Current: cyan pulse
- Cleared: vehicle filled + CLEARED stamp
- Elite (10): neon frame

License-plate UI can remain for the **Algebra Licenses wing** later; Precedence home = garage.

3.4 Modes

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Mode · Clock · Vehicle uses · Purpose
|---|---|---|---|
Practice · No · Cosmetic only (idle pose) · Learn the new rule
Rally · Yes · Full: speed, hubcaps, engine · Earn km + feel rank
Forks / Race signs · Yes · Same · Later polish; 2→4→8 lanes

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4. Mapping to the current codebase

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Concept-art era · Closest `engine.js` level today · Gap to close
|---|---|---|
1 On foot · L1 Sums (+, brackets) · Rename; foot sprite + mesa world
2 Litter · L3 Products (adds x) – *order differs* · Reorder so x comes before "hard +- drills", or keep
L2 as Differences *without* calling it a new vehicle era
3 Horse · L2-ish brackets emphasis · Dedicated bracket-trap generator
4 Carriage · L4 Quotients · ÷ world
5 Model T · L4 hard mix / long · Road test gate
6 Streamliner · L5 Powers (`POWERS` off) · Turn powers on for era ≥6
7 Cruiser · L5/L6 power+bracket · Templates
8 Wedge · L6 nested · Fantasy world pack
9 EV · *missing* · Negatives in tree + generator
10 Robotaxi · *missing* / Fractions app · Fraction bars **or** deep-link to Fractions "Bakery Lane" as
sister app

```

Recommended MVP build order

1. Remap 4 live levels → Eras 1–4 vehicles + 4 world packs (foot → carriage).
2. Garage UI + unlock saves.
3. Eras 5–8 (powers on, nested, Miami).
4. Eras 9–10 (negatives, fraction bars / elite).
5. GitHub Pages ship anytime after step 2 (even with later eras "Coming soon").

5. Product thoughts (Sensei's take)

What's strong

- Fantasy ladder is memorable and maps cleanly onto "one new rule per era."

- Starting on foot fixes the current “you’re already in an 80s car” disconnect.
- Miami as Era 8 is the right dopamine spike — same role as a mid-game legendary weapon.
- Equivalence-friendly engine already matches “drive well, don’t grind arithmetic.”

What to watch

- **Ten eras vs six engine levels:** don’t fake ten by only swapping skins. Each era needs a **felt** rule change or trap set.
- **Cleopatra litter:** concept art is fun; keep it classy (royal procession, not costume joke). Audience was locked 16+ for Algebra Licenses; if Precedence also aims younger later, litter copy may need a softer “palanquin / royal ride” label.
- **Scope creep into Algebra Licenses:** plates 2–11 are a whole second game. For MVP, **Precedence = eras 1–10 only**. Fractions stays its own Bakery app; robotaxi can **tease** fractions without swallowing that ladder.
- **Chronology pedants:** shrug in the About blurb (“travel rank, not a history exam”). Math History 101 is the museum; Precedence is the arcade.

GitHub

Still trivial: `static index.html`, `localStorage`, `Pages`. Title/rename Algebra Licenses → Precedence when we push.

6. Open questions for Julius

1. **Scope of v1 Pages ship:** Eras 1–4 playable + 5–10 locked postcards, or wait until Miami (Era 8) exists?
2. **Stamp rule:** keep 10 clean → stamp, 5 stamps → clear? Or lighter for Precedence (e.g. 3 stamps)?
3. **Must Rally be required to unlock the next vehicle, or can Practice alone clear eras?** (I lean Practice-can-clear, Rally-for-glory.)
4. **Litter naming:** keep “Cleopatra / royal litter,” or neutral “Royal litter” only?
5. **Era 10:** build fraction bars inside Precedence, or end at Era 9 and deep-link “Continue in Fractions”?
6. **Algebra Licenses wing:** hide entirely from Precedence MVP home, or show greyed “Coming soon” garage door?
7. **Attract screen:** dusk footpath vs silent garage fly-through of **unlocked** vehicles only?
8. **Dojo gating:** originally Sums/Differences/Products/Quotients apps unlocked levels — still required, or Precedence self-contained?
9. **Age tone:** keep 16+ arcade voice, or soften for 8+ once foot eras exist?
10. **Repo name:** precedence on GitHub now?

7. Proposed lock (if you say “ship it”)

- Precedence MVP = **ten eras, vehicle = rank, Miami = Era 8**.
- Practice can clear; Rally is the showcase.
- Algebra Licenses + Fractions = sister apps, not on the Precedence home strip.
- First Pages build after Eras 1–4 + garage shell; Miami before calling it “1.0.”